AP CALCULUS AB

Unit 1 • Course notes

Limits and Continuity

Topics 1.1 to 1.16 • About 10–12% of the AP Calculus AB exam

In this unit you learn what a limit is, how to find limits from graphs, tables and formulas, and how limits define continuity and asymptotes. Limits are the foundation of derivatives (Units 2–5) and integrals (Units 6–8).

1.1 Introducing Calculus: Can Change Occur at an Instant?

The average rate of change of f on $[a, a+h]$ is $\dfrac{f(a+h)-f(a)}{h}$. It needs two points. To describe change *at an instant* ($h = 0$) we cannot divide by zero, so we look at what the average rates approach as h gets closer and closer to 0. This idea is a **limit**.

Example

A ball's height is $s(t) = 20t - 5t^2$ meters. Average velocities on $[1, 1+h]$:

h	0.1	0.01	0.001	-0.001
$\dfrac{s(1+h)-s(1)}{h}$	9.5	9.95	9.995	10.005

Solution. The average velocities approach 10, so the velocity at $t = 1$ is 10 m/s.

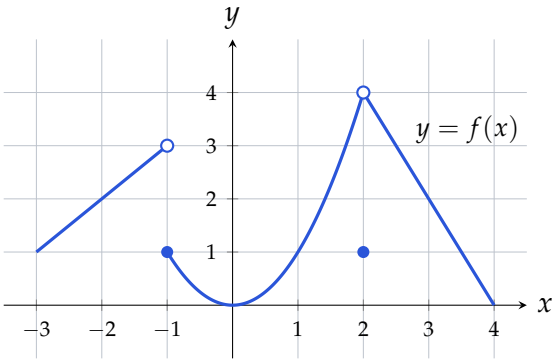
1.2 Defining Limits and Using Limit Notation

Key idea

$\lim\limits_{x\to c} f(x) = L$ means: $f(x)$ can be made as close to L as we want by taking x close enough to c (but $x \neq c$).

- $\lim\limits_{x\to c^-} f(x)$: left-hand limit ($x < c$). $\lim\limits_{x\to c^+} f(x)$: right-hand limit ($x > c$).
- $\lim\limits_{x\to c} f(x) = L$ **if and only if** both one-sided limits exist and equal L.
- The value $f(c)$ does not affect the limit; $f(c)$ may even be undefined.

1.3 Estimating Limit Values from Graphs



Example

Use the graph of f above.
Solution. At $x = -1$: $\lim_{x\to -1^-} f(x) = 3$ and $\lim_{x\to -1^+} f(x) = 1$. They differ, so $\lim_{x\to -1} f(x)$ does not exist (DNE).
At $x = 2$: both one-sided limits are 4, so $\lim_{x\to 2} f(x) = 4$, even though $f(2) = 1$.

A limit also fails to exist when f grows without bound (vertical asymptote) or oscillates (for example $\sin\frac{1}{x}$ near 0).

1.4 Estimating Limit Values from Tables

Read the outputs as x approaches c from *both* sides.

Example

x	1.9	1.99	1.999	2.001	2.01	2.1
$g(x)$	3.70	3.97	3.997	4.003	4.03	4.30

Solution. The values approach 4 from both sides, so $\lim_{x\to 2} g(x)$ appears to be 4. A table suggests a limit; it does not prove it.

1.5 Determining Limits Using Algebraic Properties of Limits

Key idea
If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then

$$\lim_{x\to c}(f \pm g) = L \pm M, \quad \lim_{x\to c}(kf) = kL, \quad \lim_{x\to c}(fg) = LM, \quad \lim_{x\to c}\frac{f}{g} = \frac{L}{M}\ (M \neq 0).$$

For polynomials, rational functions (where defined), roots, trigonometric, exponential and logarithmic functions, the limit at a point of the domain is found by **direct substitution**.

Example

$\lim_{x\to 2}\dfrac{x^2+1}{x+3} = \dfrac{5}{5} = 1.$ If $\lim_{x\to 4} f(x) = 3$ and $\lim_{x\to 4} g(x) = -2$, then $\lim_{x\to 4}\left(2f(x) - g(x)^2\right) = 6 - 4 = 2.$

1.6 Determining Limits Using Algebraic Manipulations

If substitution gives $\frac{0}{0}$, the limit is **indeterminate**: rewrite the expression first.

Factoring

$$\lim_{x\to 3}\frac{x^2-9}{x-3} = \lim_{x\to 3}\frac{(x-3)(x+3)}{x-3} = \lim_{x\to 3}(x+3) = 6.$$

Multiplying by the conjugate

$$\lim_{x\to 0}\frac{\sqrt{x+4}-2}{x}=\lim_{x\to 0}\frac{(x+4)-4}{x\left(\sqrt{x+4}+2\right)}=\lim_{x\to 0}\frac{1}{\sqrt{x+4}+2}=\frac{1}{4}.$$

Complex fraction

$$\lim_{x\to 2}\frac{\frac{1}{x}-\frac{1}{2}}{x-2}=\lim_{x\to 2}\frac{\frac{2-x}{2x}}{x-2}=\lim_{x\to 2}\frac{-1}{2x}=-\frac{1}{4}.$$

1.7 Selecting Procedures for Determining Limits

Strategy

1. Try **direct substitution**. A real number answer is the limit.
2. $\frac{0}{0}$: factor, use a conjugate, combine fractions, or use a known trig limit.
3. $\frac{k}{0}$ with $k\neq 0$: the limit is infinite or does not exist; study the sign of each one-sided limit.
4. Piecewise or absolute value functions: compute both one-sided limits.

Example

$\lim_{x\to 0^-}\frac{|x|}{x}=-1$ and $\lim_{x\to 0^+}\frac{|x|}{x}=1$, so $\lim_{x\to 0}\frac{|x|}{x}$ does not exist.

1.8 Determining Limits Using the Squeeze Theorem

Squeeze theorem

If $g(x)\le f(x)\le h(x)$ for x near c, and $\lim_{x\to c}g(x)=\lim_{x\to c}h(x)=L$, then $\lim_{x\to c}f(x)=L$.

Example

Find $\lim_{x\to 0}x^2\sin\frac{1}{x}$.

Solution. Since $-1\le\sin\frac{1}{x}\le 1$, we have $-x^2\le x^2\sin\frac{1}{x}\le x^2$. Both bounds approach 0, so the limit is 0.

Two trigonometric limits to know

$$\lim_{x\to 0}\frac{\sin x}{x}=1 \qquad \lim_{x\to 0}\frac{1-\cos x}{x}=0$$

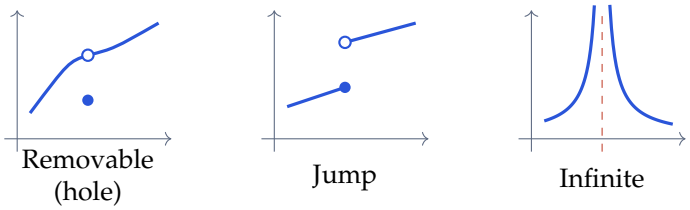
For example, $\lim_{x\to 0}\frac{\sin 3x}{x}=\lim_{x\to 0}3\cdot\frac{\sin 3x}{3x}=3.$

1.9 Connecting Multiple Representations of Limits

The same limit can be read from a graph, a table, or a formula. On the exam you may need to combine them: for example, $\lim_{x\to 2}\left(f(x)\,g(x)\right)$ where f is given by a graph and g by a formula. Find each limit separately, then apply the properties of Topic 1.5.

AP exam tip
For a composite limit $\lim_{x\to c} f(g(x))$, first find $\lim_{x\to c} g(x) = b$, then decide *from which side* $g(x)$ approaches b before reading f near b.

1.10 Exploring Types of Discontinuities



- **Removable**: $\lim_{x\to c} f(x)$ exists but is not equal to $f(c)$ (or $f(c)$ is undefined).
- **Jump**: both one-sided limits exist but are different.
- **Infinite**: at least one one-sided limit is $\pm\infty$ (vertical asymptote).

1.11 Defining Continuity at a Point

Key idea
f is **continuous at** $x = c$ if all three conditions hold:

1. $f(c)$ is defined;
2. $\lim_{x\to c} f(x)$ exists;
3. $\lim_{x\to c} f(x) = f(c)$.

AP exam tip
To justify continuity on a free-response question, state all three conditions with numbers, for example "$f(1) = 2$, $\lim_{x\to 1^-} f(x) = 2$, $\lim_{x\to 1^+} f(x) = 2$, so $\lim_{x\to 1} f(x) = f(1)$ and f is continuous at $x = 1$."

1.12 Confirming Continuity over an Interval

A function is continuous on an interval if it is continuous at every point of the interval (one-sided at endpoints). Polynomials, rational, power, exponential, logarithmic and trigonometric functions are **continuous on their domains**. Sums, products, quotients (where the denominator is nonzero) and compositions of continuous functions are continuous.

Example

$f(x) = \dfrac{\ln x}{x-3}$ is continuous on $(0,3)\cup(3,\infty)$: $\ln x$ needs $x > 0$ and the denominator is zero at $x = 3$.

1.13 Removing Discontinuities

Defining a value

$g(x) = \dfrac{x^2-4}{x-2}$ has a removable discontinuity at $x=2$ since $\lim_{x\to 2} g(x) = 4$. Defining $g(2)=4$ makes g continuous at 2.

Finding a constant

For which k is $f(x) = \begin{cases} kx+1, & x<2 \\ x^2-k, & x\geq 2 \end{cases}$ continuous at $x=2$?

Solution. Left limit: $2k+1$. Right limit and $f(2)$: $4-k$. Set them equal: $2k+1=4-k$, so $k=1$.

1.14 Connecting Infinite Limits and Vertical Asymptotes

Key idea

If $\lim\limits_{x\to c^-} f(x) = \pm\infty$ or $\lim\limits_{x\to c^+} f(x) = \pm\infty$, the line $x=c$ is a **vertical asymptote**. Saying the limit "equals ∞" describes the behavior; the limit still does not exist as a real number.

Example

$\lim\limits_{x\to 3^-} \dfrac{2}{x-3} = -\infty$ and $\lim\limits_{x\to 3^+} \dfrac{2}{x-3} = +\infty$: $x=3$ is a vertical asymptote. Also $\lim\limits_{x\to 0^+} \ln x = -\infty$ and $\lim\limits_{x\to \frac{\pi}{2}^-} \tan x = +\infty$.

1.15 Connecting Limits at Infinity and Horizontal Asymptotes

Key idea

If $\lim\limits_{x\to\infty} f(x) = L$ or $\lim\limits_{x\to-\infty} f(x) = L$, the line $y=L$ is a **horizontal asymptote**.

- Rational functions: compare the degrees (as in Precalculus).
- Growth rates as $x\to\infty$: $\ln x \ll x^p\ (p>0) \ll a^x\ (a>1)$. The fastest-growing term dominates.

Example

$\lim\limits_{x\to\infty} \dfrac{5x^2-x}{2x^2+3} = \dfrac{5}{2}$, $\lim\limits_{x\to\infty} \dfrac{x^3}{e^x} = 0$, $\lim\limits_{x\to-\infty} \dfrac{3x}{\sqrt{x^2+1}} = -3$.

For the last one, $\sqrt{x^2} = |x| = -x$ when $x<0$, so the quotient behaves like $\dfrac{3x}{-x} = -3$.

1.16 Working with the Intermediate Value Theorem (IVT)

Intermediate Value Theorem

If f is **continuous** on the closed interval $[a,b]$ and k is any number between $f(a)$ and $f(b)$, then there is at least one number c in $[a,b]$ with $f(c)=k$.

Example

Show that $f(x) = x^3 + x - 1$ has a zero in $[0, 1]$.
Solution. f is a polynomial, so it is continuous on $[0, 1]$. $f(0) = -1 < 0$ and $f(1) = 1 > 0$. Since 0 is between $f(0)$ and $f(1)$, the IVT guarantees a c in $[0, 1]$ with $f(c) = 0$.

AP exam tip

An IVT justification always names three things: the function is continuous on the closed interval, the two endpoint values, and that the target value lies between them.

Unit 1 summary

- The limit exists only if both one-sided limits exist and are equal.
- Try substitution first; $\frac{0}{0}$ means rewrite.
- $\frac{k}{0}$ ($k \neq 0$): infinite limit, check signs.
- $\lim_{x \to 0} \frac{\sin x}{x} = 1$; squeeze for bounded factors.
- Continuity: $f(c)$ defined, limit exists, they are equal.
- Discontinuities: removable, jump, infinite.
- Vertical asymptote: an infinite one-sided limit.
- Horizontal asymptote: a finite limit at $\pm\infty$.
- $\ln x \ll x^p \ll a^x$ as $x \to \infty$.
- IVT needs continuity on a closed interval.